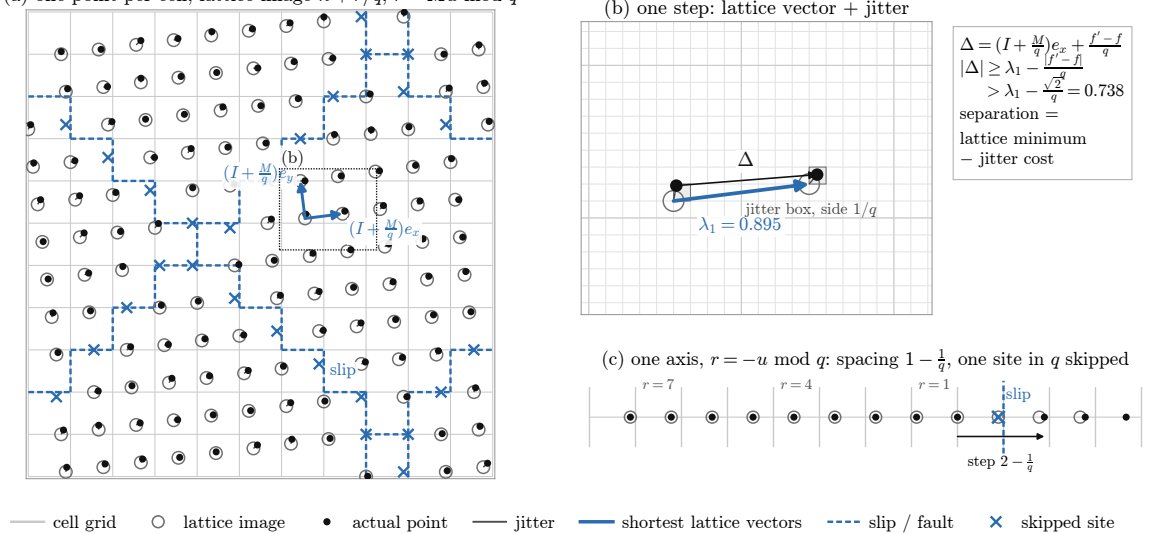
(a) one point per cell; lattice image $n + r/q$, $r = Mu \bmod q$
(b)
$(I + \frac{M}{q})e_y$
$(I - \frac{M}{q})e_x$
slip
(b) one step: lattice vector + jitter
Δ
jitter box, side $1/q$
$\lambda_1 = 0.895$
$\Delta = (I + \frac{M}{q})e_x + \frac{f' - f}{q}$
$|\Delta| \geq \lambda_1 - \frac{|f' - f|}{q}$
$> \lambda_1 - \frac{\sqrt{2}}{q} = 0.738$
separation =
lattice minimum
− jitter cost
(c) one axis, $r = -u \bmod q$: spacing $1 - \frac{1}{q}$, one site in q skipped
$r = 7$
$r = 4$
$r = 1$
slip
step $2 - \frac{1}{q}$
cell grid
lattice image
actual point
jitter
shortest lattice vectors
slip / fault
skipped site